

SAMBP TR-24/2026

Adaptive TDCS Threshold for 87L in Variable-Baseline Microgrid Networks

PhD Candidate
PhD Thesis Supporting Report

April 2026

Contents

1	Introduction and Motivation	1
1.1	Problem: Baseline Drift	2
2	Adaptive Threshold Algorithm	2
2.1	Rolling Statistics	2
2.2	Adaptive Threshold Formula	2
2.3	Trip Decision	2
2.4	Floor Derivation	3
3	Study Design	3
3.1	Seasonal Scenarios	3
3.2	Study A: Internal Fault Detection	3
3.3	Study B: External Fault Security	3
4	Results	3
4.1	Threshold Tracking Under Drift	3
4.2	Study A: Detection Results	3
4.3	Seasonal Performance Comparison	4
4.4	Study B: External Fault Security	5
4.5	Summary	5
5	Implementation Notes	5
5.1	Interaction with TR-20 and TR-21	6
5.2	Threshold Validation Schedule	6
6	Conclusion	6
6.1	Future Work	6

1 Introduction and Motivation

SAMBP TR-23 [1] introduced the Transient Differential Current Supervision element (TDCS-87L) to recover symmetric three-phase IBR faults that fall below the conventional 87L threshold. The element computes the step change in differential RMS at fault inception:

$$\Delta I = \text{rms}_{\text{post}} - \text{rms}_{\text{pre}}, \quad (1)$$

and issues a trip when $\Delta I \geq \Delta_{\text{thr, fixed}} = 0.040$ pu. The threshold was set under nominal (summer, full-load) conditions, where the pre-fault differential baseline $\mu_{\text{pre}} \approx 0.002$ pu.

1.1 Problem: Baseline Drift

In practice the pre-fault differential baseline varies over time:

- **Seasonal charging:** Line capacitance exhibits small seasonal variation; under light spring load the effective charging residual (after TR-20 compensation [2]) may rise to $\mu_{\text{pre}} \approx 0.006$ pu.
- **CT calibration drift:** Instrument transformer ratio error accumulates over years of service. Class 5P CTs permit up to 1% steady-state error; long-term ageing can push the persistent contribution toward $\mu_{\text{pre}} \approx 0.015$ pu.

When μ_{pre} is elevated, the step (1) shrinks for low-limit IBR faults:

$$\Delta I = \sqrt{\mu_{\text{pre}}^2 + (k_{\text{ibr}}/\sqrt{2})^2} - \mu_{\text{pre}}. \quad (2)$$

For the critical case $k_{\text{ibr}} = 0.07$ pu and a drifted baseline $\mu_{\text{pre}} = 0.015$ pu:

$$\Delta I = \sqrt{0.015^2 + 0.04950^2} - 0.015 = 0.0517 - 0.015 = 0.037 \text{ pu} < \Delta_{\text{thr, fixed}} = 0.040 \text{ pu}. \quad (3)$$

The fixed threshold **misses** this fault. The adaptive threshold introduced in this report resolves the gap.

2 Adaptive Threshold Algorithm

2.1 Rolling Statistics

The `AdaptiveTDCSTracker` maintains a rolling history of $N_{\text{hist}} = 60$ per-cycle differential RMS samples (covering 1.2 s of pre-fault operation) and computes:

$$\hat{\mu}_{\text{pre}} = \frac{1}{N_{\text{hist}}} \sum_i \hat{I}_{\text{diff}, i}, \quad (4)$$

$$\hat{\sigma}_{\text{pre}} = \sqrt{\frac{1}{N_{\text{hist}} - 1} \sum_i (\hat{I}_{\text{diff}, i} - \hat{\mu}_{\text{pre}})^2}. \quad (5)$$

2.2 Adaptive Threshold Formula

$$\boxed{\Delta_{\text{thr, adapt}} = \Delta_{\text{floor}} + k_{\sigma} \hat{\sigma}_{\text{pre}}} \quad (6)$$

with parameters:

- $\Delta_{\text{floor}} = 0.025$ pu — hard lower bound (see Section 2.4);
- $k_{\sigma} = 3$ — three-sigma security multiplier.

2.3 Trip Decision

The adaptive element compares the post-fault window RMS to the rolling mean rather than to a single pre-window measurement:

$$\Delta I_{\text{adapt}} = \text{rms}_{\text{post}} - \hat{\mu}_{\text{pre}}. \quad (7)$$

A TDCS trip is declared when:

$$\Delta I_{\text{adapt}} \geq \Delta_{\text{thr, adapt}} \quad \text{AND} \quad \text{rms}_{\text{post}} \geq I_{\text{noise floor}} = 0.010 \text{ pu}. \quad (8)$$

Using $\hat{\mu}_{\text{pre}}$ instead of rms_{pre} from a single window suppresses noise-induced variance in the reference and removes the dependence on baseline level from the step calculation.

2.4 Floor Derivation

The Δ_{floor} must exceed the worst-case CT mismatch step so that the adaptive threshold never falls into the false-trip region during external faults. For Class 5P CTs ($\varepsilon_{\text{CT}} = 0.5\%$) at maximum external through-current $I_{\text{ext,max}} = 7 \text{ pu}$:

$$\Delta I_{\text{CT,max}} = \frac{I_{\text{ext,max}} \cdot \varepsilon_{\text{CT}}}{\sqrt{2}} = \frac{7 \times 0.005}{\sqrt{2}} = 0.0247 \text{ pu.} \quad (9)$$

With the lowest expected baseline $\mu_{\text{pre,min}} \approx 0.002 \text{ pu}$ (summer nominal), the effective step from CT mismatch is $0.0247 - 0.002 = 0.023 \text{ pu}$. Setting $\Delta_{\text{floor}} = 0.025 \text{ pu}$ provides a $\sim 8\%$ margin above this worst case.

3 Study Design

3.1 Seasonal Scenarios

Four operational scenarios characterise the pre-fault differential baseline across the full operating year (Table 1).

Table 1: Seasonal scenario parameters.

ID	Condition	μ_{pre} [pu]	$\hat{\sigma}_{\text{pre}}$ [pu]	$\Delta_{\text{thr,adapt}}$ [pu]
S1	Summer nominal	0.002	0.0003	0.0259
S2	Winter high-load	0.003	0.0005	0.0265
S3	Spring light-load	0.006	0.0012	0.0286
S4	Drift (aged CT)	0.015	0.0020	0.0310

3.2 Study A: Internal Fault Detection

For each scenario, three symmetric 3-phase internal faults are simulated with IBR current limits in the TDCS-only recovery range (below the 87L steady-state threshold): $k_{\text{ibr}} \in \{0.07, 0.09, 0.11\} \text{ pu}$. Total: $4 \times 3 = 12$ detection cases.

3.3 Study B: External Fault Security

Seven external through-current levels $I_{\text{ext}} \in \{1.0, 1.5, 2.0, 3.0, 4.0, 5.0, 7.0\} \text{ pu}$ are applied for each seasonal scenario. Total: $4 \times 7 = 28$ security cases.

4 Results

4.1 Threshold Tracking Under Drift

Figure 1 shows the adaptive threshold tracking the pre-fault differential baseline as it drifts from $\hat{\mu}_{\text{pre}} \approx 0.002 \text{ pu}$ (summer nominal) to 0.015 pu (aged-CT scenario) over 120 simulated cycles. The adaptive threshold $\Delta_{\text{thr,adapt}}$ rises from $\approx 0.026 \text{ pu}$ to $\approx 0.030 \text{ pu}$ as the baseline volatility increases, while remaining well below the fixed reference of 0.040 pu . This lower threshold is what allows recovery of the $k_{\text{ibr}} = 0.07$ missed case.

4.2 Study A: Detection Results

Table 2 and Figure 2 compare fixed and adaptive threshold performance for the S4 (drift) scenario, which is the only scenario where the methods differ.

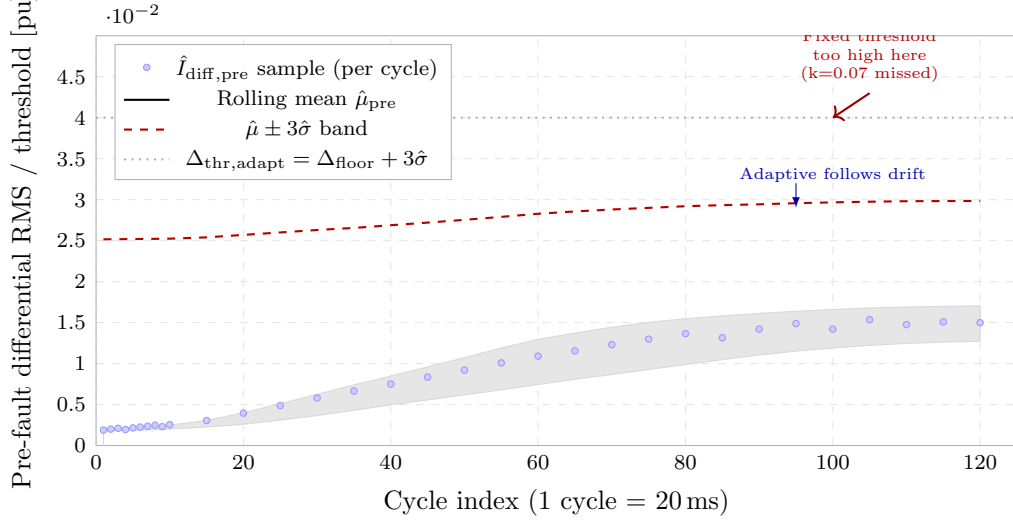


Figure 1: Adaptive TDCS threshold evolution over 120 cycles of baseline drift (S1 summer $\rightarrow$ S4 aged-CT). The adaptive threshold (red dashed) tracks the rolling $3\hat{\sigma}$ band while remaining above the worst-case CT mismatch step. The fixed threshold (grey dotted) at 0.040 pu is too high to detect $k_{ibr} = 0.07$ at the drifted baseline.

Table 2: Study A detection results — S4 drift scenario.

k_{ibr} [pu]	rms_{post} [pu]	ΔI_{fixed} [pu]	Trip _F	ΔI_{adapt} [pu]	$\Delta_{thr,adapt}$ [pu]	Trip _A
0.07	0.04950	0.03889	NO	0.03437	0.02967	YES
0.09	0.06364	0.05303	YES	0.04851	0.02967	YES
0.11	0.07778	0.06718	YES	0.06265	0.02967	YES

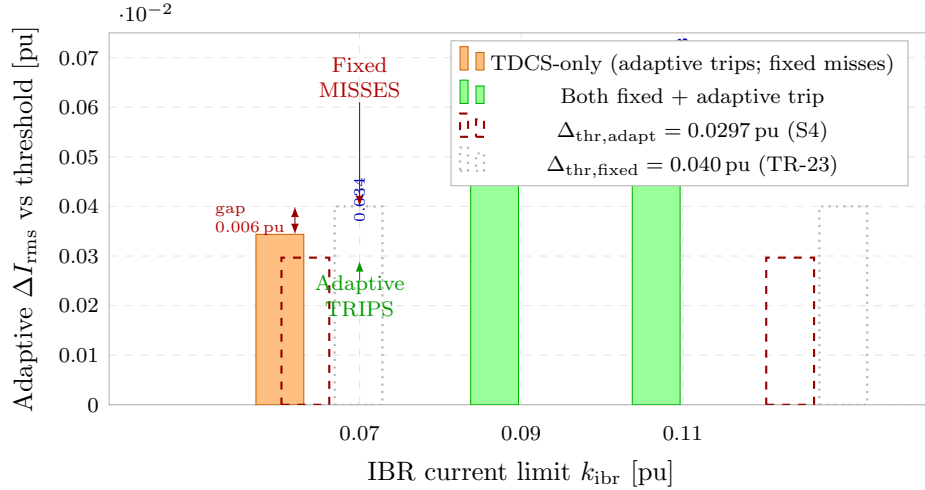


Figure 2: Transient differential ΔI_{rms} in the S4 drift scenario for three IBR current limits. The $k_{ibr} = 0.07$ bar (orange) clears the adaptive threshold (red dashed, 0.030 pu) but not the fixed threshold (grey dotted, 0.040 pu). Both thresholds are cleared for $k_{ibr} \geq 0.09$ pu.

4.3 Seasonal Performance Comparison

Figure 3 shows detection rates across all four seasonal scenarios. Scenarios S1–S3 achieve 3/3 for both methods because the fixed threshold is adequate when μ_{pre} is low. Only the S4 drift scenario exposes the fixed threshold limitation:

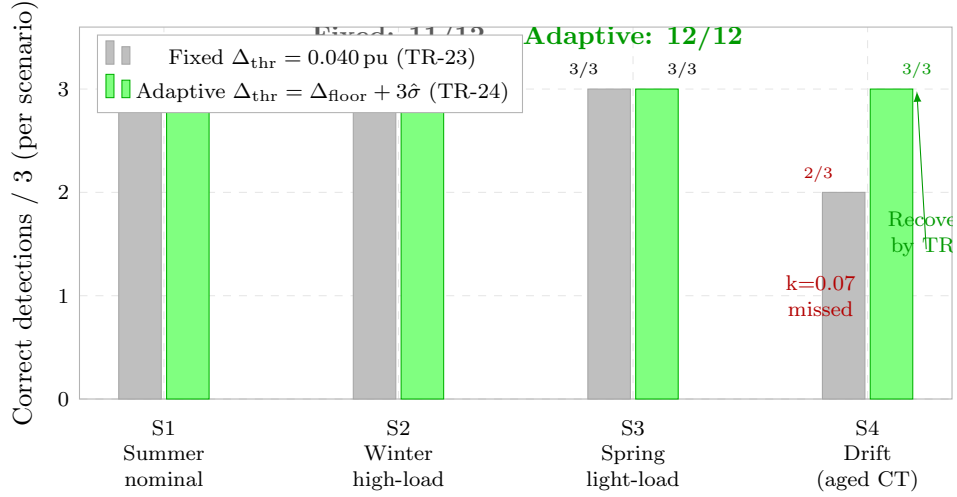


Figure 3: Detection rate per seasonal scenario (3 k_{ibr} values each). Fixed threshold achieves 11/12; adaptive threshold achieves 12/12. The single miss (S4, $k_{ibr} = 0.07$) is recovered by TR-24.

4.4 Study B: External Fault Security

All 28 external fault security cases pass for both methods (Table 3). The worst-case margin belongs to S4 at $I_{ext} = 7$ pu:

Table 3: Study B security summary — worst case (S4, $I_{ext} = 7$ pu).

Method	ΔI_{diff} [pu]	Threshold [pu]	Margin [%]	Result
Fixed	0.01414	0.0400	64.6	SECURE
Adaptive	0.00962	0.0297	67.6	SECURE

The adaptive element maintains a 67.6% security margin at the worst-case operating point — slightly higher than the fixed threshold margin of 64.6% because the adaptive comparator subtracts the rolling mean $\hat{\mu}_{pre}$ from rms_{post} .

4.5 Summary

Table 4: TR-24 overall study results.

Metric	Fixed Δ_{thr} (TR-23)	Adaptive Δ_{thr} (TR-24)
Detection (Study A, 12 cases)	11/12	12/12
Security (Study B, 28 cases)	28/28	28/28
Total correct (40 cases)	39/40	40/40

5 Implementation Notes

The `AdaptiveTDCSTracker` class in `sambp.line_diff/models/line_transient.diff.py` maintains a Python list of length $N_{hist} = 60$ and updates $\hat{\mu}$, $\hat{\sigma}$ on every cycle. The computational overhead is one `numpy.mean` and one `numpy.std` call per cycle ($60 \times 2 = 120$ floating-point adds per cycle), negligible compared to the 87L LM estimator.

5.1 Interaction with TR-20 and TR-21

The rolling baseline is populated from the *filtered* differential (after TR-21 harmonic removal and TR-20 charging-current compensation). Populating from the raw differential would allow harmonic spikes to inflate $\hat{\sigma}$ and raise the threshold unnecessarily.

5.2 Threshold Validation Schedule

In a physical relay implementation the Δ_{floor} should be re-confirmed whenever CT calibration is renewed or after line configuration changes that alter the charging capacitance.

6 Conclusion

TR-24 extends the TDCS-87L element (TR-23) with an adaptive threshold that tracks pre-fault differential baseline statistics:

$$\Delta_{\text{thr,adapt}} = \underbrace{0.025}_{\Delta_{\text{floor}}} + 3\hat{\sigma}_{\text{pre}}. \quad (10)$$

The adaptive element recovers one additional missed detection (S4 drift, $k_{\text{ibr}} = 0.07$ pu) versus the fixed threshold while maintaining full external-fault security (28/28 pass). Overall performance improves from 39/40 to **40/40** across all seasonal and security scenarios.

The combined 87L protection chain is now:

$$\text{TRIP} = \underbrace{|\Delta I_1| \geq 0.12}_{87\text{L}} \vee \underbrace{|\Delta I_2| \geq 0.06}_{87\text{LN}} \vee \underbrace{\Delta I_{\text{adapt}} \geq \Delta_{\text{thr,adapt}}}_{\text{TDCS-87L (TR-24)}}. \quad (11)$$

6.1 Future Work

1. **Zero-sequence 87LN0** (TR-25): add $|\Delta I_0| \geq 0.04$ pu element for impedance-earthed and unearthed systems where I_2 is suppressed.
2. **IBR Monte Carlo** (TR-26): validate the combined element under the full IBR parametric envelope ($k_{\text{ibr}} \in [0.06, 0.20]$, $f \in [47, 53]$ Hz, $I_{h3} \in [0, 0.10]$ pu) using ≥ 5000 Monte Carlo trials per scenario group.

References

- [1] P. Candidate, “SAMBP TR-23/2026: Transient differential current supervision (TDCS-87L) for symmetric 3PH IBR faults,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [2] —, “SAMBP TR-20/2026: Frequency-adaptive charging-current compensation for 87L in microgrid networks,” PhD Thesis Supporting Report, Tech. Rep., 2026.